
Workgroup: Network Working Group
Internet-Draft: draft-whelan-chio-protocol-00
Published: 5 October 2026
Intended: Standards Track
Status:
Expires: 8 April 2027
Author: C. Whelan
Backbay Industries

The Chio Protocol: Signed Capabilities and Receipts for Mediated Tool Execution

Abstract

This document is being written. The abstract replaces this sentence.

The RFC Editor will remove this note

About This Document

This note is to be removed before publishing as an RFC.

Status information for this document may be found at <https://datatracker.ietf.org/doc/draft-whelan-chio-protocol/>.

Discussion of this document takes place on the agentproto Working Group mailing list (<mailto:agentproto@ietf.org>), which is archived at <https://mailarchive.ietf.org/arch/browse/agentproto/>. Subscribe at <https://www.ietf.org/mailman/listinfo/agentproto/>.

Source for this draft and an issue tracker can be found at <https://github.com/backbay-labs/chio>.

Status of This Memo

This Internet-Draft is submitted in full conformance with the provisions of BCP 78 and BCP 79.

Internet-Drafts are working documents of the Internet Engineering Task Force (IETF). Note that other groups may also distribute working documents as Internet-Drafts. The list of current Internet-Drafts is at <https://datatracker.ietf.org/drafts/current/>.

Internet-Drafts are draft documents valid for a maximum of six months and may be updated, replaced, or obsoleted by other documents at any time. It is inappropriate to use Internet-Drafts as reference material or to cite them other than as "work in progress."

This Internet-Draft will expire on 8 April 2027.

Copyright Notice

Copyright (c) 2026 IETF Trust and the persons identified as the document authors. All rights reserved.

This document is subject to BCP 78 and the IETF Trust's Legal Provisions Relating to IETF Documents (<https://trustee.ietf.org/license-info>) in effect on the date of publication of this document. Please review these documents carefully, as they describe your rights and restrictions with respect to this document. Code Components extracted from this document must include Revised BSD License text as described in Section 4.e of the Trust Legal Provisions and are provided without warranty as described in the Revised BSD License.

Table of Contents

1. Introduction	3
2. Conventions and Terminology	3
3. Protocol Overview	3
4. Encoding and Cryptography	3
5. Capability Tokens	3
6. Receipts	3
7. Budgets and Metering	3
8. Governed Transactions	4
9. Receipt Checkpoints	4
10. Native Transport	4
11. Hosted MCP Binding	4
12. Trust-Control Interface	4
13. Versioning and Negotiation	4
14. Error Model	4
15. Security Considerations	4
16. Privacy Considerations	4
17. IANA Considerations	4
18. Implementation Status	4
19. References	5
19.1. Normative References	5
19.2. Informative References	6

Appendix A. Examples	7
Appendix B. Test Vectors	8
Appendix C. Relationship to Other Work	8
Appendix D. Document History	8
Acknowledgments	8
Author's Address	8

1. Introduction

This section is being written.

2. Conventions and Terminology

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 [[RFC2119](#)] [[RFC8174](#)] when, and only when, they appear in all capitals, as shown here.

3. Protocol Overview

This section is being written.

4. Encoding and Cryptography

This section is being written.

5. Capability Tokens

This section is being written.

6. Receipts

This section is being written.

7. Budgets and Metering

This section is being written.

8. Governed Transactions

This section is being written.

9. Receipt Checkpoints

This section is being written.

10. Native Transport

This section is being written.

11. Hosted MCP Binding

This section is being written.

12. Trust-Control Interface

This section is being written.

13. Versioning and Negotiation

This section is being written.

14. Error Model

This section is being written.

15. Security Considerations

This section is being written.

16. Privacy Considerations

This section is being written.

17. IANA Considerations

This section is being written.

18. Implementation Status

This section is to be removed before publishing as an RFC.

This section is being written.

19. References

19.1. Normative References

- [FIPS180-4]** National Institute of Standards and Technology, "Secure Hash Standard (SHS)", FIPS PUB 180-4, DOI 10.6028/NIST.FIPS.180-4, August 2015, <<https://doi.org/10.6028/NIST.FIPS.180-4>>.
- [FIPS186-5]** National Institute of Standards and Technology, "Digital Signature Standard (DSS)", FIPS PUB 186-5, DOI 10.6028/NIST.FIPS.186-5, February 2023, <<https://doi.org/10.6028/NIST.FIPS.186-5>>.
- [FIPS204]** National Institute of Standards and Technology, "Module-Lattice-Based Digital Signature Standard", FIPS PUB 204, DOI 10.6028/NIST.FIPS.204, August 2024, <<https://doi.org/10.6028/NIST.FIPS.204>>.
- [ISO4217]** International Organization for Standardization, "Codes for the representation of currencies", ISO 4217:2015, August 2015, <<https://www.iso.org/iso-4217-currency-codes.html>>.
- [JSON-RPC]** JSON-RPC Working Group, "JSON-RPC 2.0 Specification", January 2013, <<https://www.jsonrpc.org/specification>>.
- [MCP]** Model Context Protocol, "Model Context Protocol Specification, Version 2025-11-25", 25 November 2025, <<https://modelcontextprotocol.io/specification/2025-11-25>>.
- [RFC2119]** Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997, <<https://www.rfc-editor.org/rfc/rfc2119>>.
- [RFC3279]** Bassham, L., Polk, W., and R. Housley, "Algorithms and Identifiers for the Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile", RFC 3279, DOI 10.17487/RFC3279, May 2002, <<https://www.rfc-editor.org/rfc/rfc3279>>.
- [RFC4648]** Josefsson, S., "The Base16, Base32, and Base64 Data Encodings", RFC 4648, DOI 10.17487/RFC4648, October 2006, <<https://www.rfc-editor.org/rfc/rfc4648>>.
- [RFC5234]** Crocker, D., Ed. and P. Overell, "Augmented BNF for Syntax Specifications: ABNF", STD 68, RFC 5234, DOI 10.17487/RFC5234, January 2008, <<https://www.rfc-editor.org/rfc/rfc5234>>.
- [RFC7405]** Kyzivat, P., "Case-Sensitive String Support in ABNF", RFC 7405, DOI 10.17487/RFC7405, December 2014, <<https://www.rfc-editor.org/rfc/rfc7405>>.
- [RFC8032]** Josefsson, S. and I. Liusvaara, "Edwards-Curve Digital Signature Algorithm (EdDSA)", RFC 8032, DOI 10.17487/RFC8032, January 2017, <<https://www.rfc-editor.org/rfc/rfc8032>>.

- [RFC8126] Cotton, M., Leiba, B., and T. Narten, "Guidelines for Writing an IANA Considerations Section in RFCs", BCP 26, RFC 8126, DOI 10.17487/RFC8126, June 2017, <<https://www.rfc-editor.org/rfc/rfc8126>>.
- [RFC8174] Leiba, B., "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words", BCP 14, RFC 8174, DOI 10.17487/RFC8174, May 2017, <<https://www.rfc-editor.org/rfc/rfc8174>>.
- [RFC8259] Bray, T., Ed., "The JavaScript Object Notation (JSON) Data Interchange Format", STD 90, RFC 8259, DOI 10.17487/RFC8259, December 2017, <<https://www.rfc-editor.org/rfc/rfc8259>>.
- [RFC8446] Rescorla, E., "The Transport Layer Security (TLS) Protocol Version 1.3", RFC 8446, DOI 10.17487/RFC8446, August 2018, <<https://www.rfc-editor.org/rfc/rfc8446>>.
- [RFC8785] Rundgren, A., Jordan, B., and S. Erdtman, "JSON Canonicalization Scheme (JCS)", RFC 8785, DOI 10.17487/RFC8785, June 2020, <<https://www.rfc-editor.org/rfc/rfc8785>>.
- [RFC9110] Fielding, R., Ed., Nottingham, M., Ed., and J. Reschke, Ed., "HTTP Semantics", STD 97, RFC 9110, DOI 10.17487/RFC9110, June 2022, <<https://www.rfc-editor.org/rfc/rfc9110>>.
- [RFC9162] Laurie, B., Messeri, E., and R. Stradling, "Certificate Transparency Version 2.0", RFC 9162, DOI 10.17487/RFC9162, December 2021, <<https://www.rfc-editor.org/rfc/rfc9162>>.
- [RFC9449] Fett, D., Campbell, B., Bradley, J., Lodderstedt, T., Jones, M., and D. Waite, "OAuth 2.0 Demonstrating Proof of Possession (DPoP)", RFC 9449, DOI 10.17487/RFC9449, September 2023, <<https://www.rfc-editor.org/rfc/rfc9449>>.
- [SEC1] Standards for Efficient Cryptography Group, "SEC 1: Elliptic Curve Cryptography, Version 2.0", May 2009, <<https://www.secg.org/sec1-v2.pdf>>.
- [SSE] WHATWG, "HTML Living Standard, Section 9.2: Server-sent events", <<https://html.spec.whatwg.org/multipage/server-sent-events.html>>.

19.2. Informative References

- [A2A] Linux Foundation A2A Project, "Agent2Agent (A2A) Protocol Specification", <<https://a2a-protocol.org/latest/specification/>>.
- [BISCUIT] Eclipse Biscuit, "Biscuit, a bearer token with offline attestation and decentralized verification", <<https://doc.biscuitsec.org/reference/specifications>>.
- [I-D.ietf-scitt-architecture] Birkholz, H., Delignat-Lavaud, A., Fournet, C., Deshpande, Y., and S. Lasker, "An Architecture for Trustworthy and Transparent Digital Supply Chains", Work in Progress, Internet-Draft, draft-ietf-scitt-architecture-22, 10 October 2025, <<https://datatracker.ietf.org/doc/html/draft-ietf-scitt-architecture-22>>.

- [I-D.ietf-wimse-arch]** Salowey, J. A., Rosomakho, Y., and H. Tschofenig, "Workload Identity in a Multi System Environment (WIMSE) Architecture", Work in Progress, Internet-Draft, draft-ietf-wimse-arch-08, 6 July 2026, <<https://datatracker.ietf.org/doc/html/draft-ietf-wimse-arch-08>>.
- [MACAROONS]** Birgisson, A., Politz, J. G., Erlingsson, U., Taly, A., Vrabie, M., and M. Lentczner, "Macaroons: Cookies with Contextual Caveats for Decentralized Authorization in the Cloud", Network and Distributed System Security Symposium NDSS 2014, February 2014, <<https://www.ndss-symposium.org/ndss2014/programme/macaroons-cookies-contextual-caveats-decentralized-authorization-cloud/>>.
- [OID4VCI]** OpenID Foundation, "OpenID for Verifiable Credential Issuance 1.0", 2025, <https://openid.net/specs/openid-4-verifiable-credential-issuance-1_0.html>.
- [OID4VP]** OpenID Foundation, "OpenID for Verifiable Presentations 1.0", 2025, <https://openid.net/specs/openid-4-verifiable-presentations-1_0.html>.
- [RFC7942]** Sheffer, Y. and A. Farrel, "Improving Awareness of Running Code: The Implementation Status Section", BCP 205, RFC 7942, DOI 10.17487/RFC7942, July 2016, <<https://www.rfc-editor.org/rfc/rfc7942>>.
- [RFC8693]** Jones, M., Nadalin, A., Campbell, B., Ed., Bradley, J., and C. Mortimore, "OAuth 2.0 Token Exchange", RFC 8693, DOI 10.17487/RFC8693, January 2020, <<https://www.rfc-editor.org/rfc/rfc8693>>.
- [RFC8792]** Watsen, K., Auerswald, E., Farrel, A., and Q. Wu, "Handling Long Lines in Content of Internet-Drafts and RFCs", RFC 8792, DOI 10.17487/RFC8792, June 2020, <<https://www.rfc-editor.org/rfc/rfc8792>>.
- [RFC9334]** Birkholz, H., Thaler, D., Richardson, M., Smith, N., and W. Pan, "Remote Attestation procedureS (RATS) Architecture", RFC 9334, DOI 10.17487/RFC9334, January 2023, <<https://www.rfc-editor.org/rfc/rfc9334>>.
- [RFC9396]** Lodderstedt, T., Richer, J., and B. Campbell, "OAuth 2.0 Rich Authorization Requests", RFC 9396, DOI 10.17487/RFC9396, May 2023, <<https://www.rfc-editor.org/rfc/rfc9396>>.
- [RFC9635]** Richer, J., Ed. and F. Imbault, "Grant Negotiation and Authorization Protocol (GNAP)", RFC 9635, DOI 10.17487/RFC9635, October 2024, <<https://www.rfc-editor.org/rfc/rfc9635>>.
- [UCAN]** UCAN Working Group, "User Controlled Authorization Network (UCAN) Specification", <<https://github.com/ucan-wg/spec>>.
- [VC-DATA-MODEL-2.0]** World Wide Web Consortium, "Verifiable Credentials Data Model v2.0", 15 May 2025, <<https://www.w3.org/TR/vc-data-model-2.0/>>.

Appendix A. Examples

This appendix is being written.

Appendix B. Test Vectors

This appendix is being written.

Appendix C. Relationship to Other Work

This appendix is being written.

Appendix D. Document History

This section is to be removed before publishing as an RFC.

This appendix is being written.

Acknowledgments

This section is being written.

Author's Address

Connor Whelan

Backbay Industries

Email: connor@backbay.io